Project 3: Problem Description

MATP32 Harmonic Analysis

3. Weighted inequalities for maximal function

Let w be a locally integrable nonnegative function on $\mathbb{R}^d$ (such a function is called a weight), let $p \in (1, \infty)$ and let

$$L^p_w(\mathbb{R}^d) = \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} w(x)|f(x)|^p \, dx < \infty \right\}.$$

Recall that the (unweighted) Hardy-Littlewood maximal function, given by

$$Mf(x) = \sup_{Q \text{ cube}: x \in Q} \frac{1}{|Q|} \int_Q |f(y)| \, dy,$$

is a bounded sublinear operator on $L^p(\mathbb{R}^d)$. The question we want to pursue in this project is for which weights w, M is also a bounded sublinear operator on $L^p_w(\mathbb{R}^d)$. Muckenhoupt proved in 1972 [23] that this is the class of weights characterised by the so-called Muckenhoupt A_p characteristic,

$$[w]_{A_p} := \sup_{Q \text{ cube}} \frac{1}{|Q|} \int_Q w(x) \, dx \, \frac{1}{|Q|} \int_Q w^{-1/(p-1)}(x) \, dx < \infty.$$

Lerner gave a much shorter proof in [21], Theorem C. Here, he even proved a precise estimate on how $\|M\|_{L^p_w \to L^p_w}$ depends on the characteristic $[w]_{A_p}$ (this was formerly proven by S. Buckley). The goal of the project is to present a proof of Theorem C in [21].

Suggested literature.

- A. K. Lerner, "An elementary approach to several results on the Hardy-Littlewood maximal operator", Proc. Amer. Math. Soc. **136** (2008), no. 8, 2829–2833.

Marking criteria for the projects in MATP32

The length of the project should be 6-12 pages in LaTeX. **Total score: 40 points**

Written part: 20 p

5p Introduction: Connection with the contents of the course (if applicable), context, motivation

5p Structure and coherence of the whole text, avoidance of notational or other inconsistencies (including typos and minor errors).

10p A clear and rigorous presentation of the proof of the main result.

Feedback: 5 hp

5p Meaningful and constructive feedback to your writing partner (about half a page)

Oral Presentation: 15 hp

5p Structure and coherence of the presentation

5p Clarity of exposition/discussion

5p Contact with audience, answer to questions

References

[21] A. K. Lerner, "An elementary approach to several results on the Hardy-Littlewood maximal operator", Proc. Amer. Math. Soc. **136** (2008), no. 8, 2829–2833.

[23] B. Muckenhoupt, *Weighted norm inequalities for the Hardy maximal function*, Trans. Amer. Math. Soc. **165** (1972), 207–226.